# Scott Powers

saberpowers.com

## Academic Experience

**Rice University** 2023–
Assistant Professor, Sport Analytics 2023–
Assistant Professor, Statistics (joint appointment) 2023–
Director, Hutchinson Leadership Initiative in Sport Analytics 2026–

## Sports Experience

### Full-Time

**Houston Astros** 2022
Assistant General Manager

**Los Angeles Dodgers** 2017–2021
Director, Quantitative Analysis 2019–2021
Senior Analyst, Research & Development 2017–2018

### Part-Time/Consulting

**Washington Nationals** 2026
Consultant

**St Louis Cardinals** 2025
Consultant

**Teamworks** (formerly Zelus Analytics) 2020, 2023–2024
Staff Data Scientist 2023–2024
Senior Data Scientist 2020

**AZ Alkmaar** 2015–2020
Data Analyst

**Oakland Athletics** 2015–2016
Analytics Consultant

## Education

**Stanford University** 2017
Ph.D. Statistics
Advisor: Rob Tibshirani

**University of Chicago** 2013
M.S. Statistics

**University of North Carolina at Chapel Hill** 2011
B.S. Mathematical Decision Sciences and Mathematics
*Highest Honors from the Department and Highest Distinction from the University*

# Research

## Working Papers

**Powers**, Graves R, Vannucci M, Buffi JH, Barrueco D, D'Angelo J, Glazer AK
"A large-scale study of pitch-tracking risk factors for professional baseball pitcher injuries"

**Powers**, Ramani S, Hahn J, Schaefer AJ
"Lead distance under a pickoff limit in Major League Baseball: A sequential game model"
*arXiv:2601.15608*

Zhu R, Love D, **Powers**
"Ball path curvature and in-game free throw shooting proficiency in the National Basketball Association"
*arXiv:2507.01238*

**Powers**, Iglesias V
"Pitch trajectory density estimation for predicting future outcomes"

## Journal Articles

**Powers**, Yurko R (2026)
"Swinging, Fast and Slow: Interpreting variation in baseball swing tracking metrics"
*The American Statistician* **80**(4) 535-549

Millington E, **Powers** (2026)
"Models in the War Room: Translating random forest predictions into player comps"
*Journal of Sports Analytics* **12** 22150218261472166

**Powers**, Stancil L, Consiglio N (2025)
"Estimating individual contributions to team success in women's college volleyball"
*Journal of Quantitative Analysis in Sports* **21**(2) 117–135

Burton T, **Powers**, Burns C, Conway G, Leach F, Senecal K (2023)
"A data-driven greenhouse gas emission rate analysis for vehicle comparisons"
*SAE International Journal of Electrified Vehicles* **12**(1)

**Powers**, McGuire V, Bernstein L, Canchola AJ, Whittemore AS (2019) "Evaluating disease prediction models using a cohort whose covariate distribution differs from that of the target population"
*Statistical Methods in Medical Research* **28**(1) 309–320

**Powers**, Hastie T, Tibshirani R (2018)
"Nuclear penalized multinomial regression with an application to predicting at bat outcomes in baseball"
*Statistical Modelling* **18**(5–6) 388–410

**Powers**, Qian J, Jung K, Schuler A, Shah NH, Hastie T, Tibshirani R (2018)
"Some methods for heterogeneous treatment effect estimation in high dimensions"
*Statistics in Medicine* **37**(11) 1767–1787

McGinnis L, **Powers**, Bangs D, Cherry A, Tibshirani R, Natkunam Y (2016)
"Double-hit diffuse large B-cell lymphomas with MYC gene rearrangements more commonly involve BCL2 than BCL6 gene rearrangements as the second hit: A large scale single institution study"
*Laboratory Investigation* **96** 362A

Burton T, **Powers** (2015)

"A linear model for estimating optimal service fraction in volleyball"
*Journal of Quantitative Analysis in Sports* **11** 117–129

**Powers**, Hastie T, Tibshirani R (2015)
"Customized training with an application to mass spectrometric imaging of cancer tissue"
*The Annals of Applied Statistics* **9** 1709–1725

**Powers**, DeJongh M, Best AA, Tintle NL (2015)
"Cautions about the reliability of pairwise gene correlations based on expression data"
*Frontiers in Microbiology* **6** 650

**Powers**, Gopalakrishnan S, Tintle NL (2011)
"Assessing the impact of non-differential genotyping errors on rare variant tests of association"
*Human Heredity* **72** 153–160

Saavedra S, **Powers**, McCotter T, Porter MA, Mucha PJ (2010)
"Mutually-antagonistic interactions in baseball networks"
*Physica A: Statistical Mechanics and its Applications* **389** 1131–1141

## Invited Conference Talks

"Safer, Not Slower: Actionable insights from a large-scale survival analysis of baseball pitching mechanics"
Conference on Complex Systems (Complexity & Sports Satellite) Binghamton, NY 2026
Joint Statistical Meetings Boston 2026

"Winning baseball games by solving statistics puzzles"
International Conference on Statistics and Data Science Vancouver 2025

"Estimating individual contributions to team success in women's college volleyball"
MIT Sloan Sports Analytics Conference Boston 2024
New England Symposium on Statistics in Sports Boston 2023

## Invited Department Seminars

"Winning baseball games by solving statistics puzzles"
Northeastern University, Network Science Institute Mar 2026
Texas A&M University, Institute of Data Science Feb 2026
Southern Methodist University, Department of Statistics and Data Science Sep 2025
Johns Hopkins University, Department of Electrical and Computer Engineering Sep 2025

"Player evaluation based on game performance"
New York Knicks (National Basketball Association) Nov 2024
TSG 1899 Hoffenheim (Bundesliga) Apr 2024

"Nuclear penalized multinomial regression"
Rice University, Department of Statistics Apr 2023

## Invited Conference Panels

"Building the Academic Playbook for Sports Analytics Education"
MIT Sloan Sports Analytics Conference Boston 2026

"The Past, Present, and Future of Sports Analytics in Industry and the Academy"
Joint Statistical Meetings Nashville 2025

"Data Science and Analytics in Sports Performance and Management"
Columbia Symposium on AI and Sports — New York 2024

## Contributed Conference Talks

"Safer, Not Slower: Actionable insights from a large-scale survival analysis of baseball pitching mechanics"
Saberseminar — Chicago 2026

"Swinging, Fast and Slow: Untangling intention and timing error from bat speed and swing length"
Cascadia Symposium on Statistics in Sports — Vancouver 2024
Saberseminar — Chicago 2024

"Pitch trajectory density estimation for predicting future outcomes"
Conference of Texas Statisticians — Houston 2024
Saberseminar — Chicago 2023

"Jointly predicting exit velocity and launch angle for batter-pitcher matchups"
Saberseminar — Boston 2016

"Nuclear penalized multinomial regression for predicting at bat outcomes in baseball"
Joint Statistical Meetings — Chicago 2016

"True wOBA: Estimation of true talent level for batters"
SABR Analytics Conference — Phoenix 2016

"Rewarding batters for baserunner advancement: A ridge-regressed Rasch model"
Saberseminar — Boston 2015

## Student Conference Talks

"Bayesian hierarchical change point model for pitcher biomechanics"
Rose Graves @ Joint Statistical Meetings — Boston 2026

"A comps-based approach for interpreting tree-based predictions with an application to the NFL draft"
Elisabeth Millington @ Carnegie Mellon Sports Analytics Conference — Pittsburgh 2025

"Ball path curvature and in-game free throw shooting proficiency in the National Basketball Association"
Ruoqian (Judy) Zhu @ New England Symposium on Statistics in Sports — Boston 2025

"The Two-Foot Rule: A game theoretic analysis of the pickoff limit in Major League Baseball"
Jacob Hahn @ Cascadia Symposium on Statistics in Sports — Vancouver 2024
Jacob Hahn @ Saberseminar — Chicago 2024

"Modifying $k$-means clustering to optimize positioning in volleyball"
Naomi Consiglio @ Women in Sports Data Symposium — Philadelphia 2024

"Do we learn more about AAA batters when they face better pitches?"
Jeff Brover @ Saberseminar — Chicago 2024

"Time warping for clustering pitcher deliveries"
Drew Haugen @ Saberseminar — Chicago 2024

### Student Conference Posters

"A comps-based approach for interpreting tree-based predictions with an application to the NFL draft"
Elisabeth Millington @ New England Symposium on Statistics in Sports — Boston 2025

"A statistical approach to sport climbing difficulty and progression"
Elizabeth Sepúvelda @ Cascadia Symposium on Statistics in Sports — Vancouver 2024

"Not All Features Are Created Equal: Player clustering and evaluation"
Andrew Kang @ Opta Forum — London 2024

## Funding

### External

**Major League Baseball Research Gift**
"Identifying Biomechanical Risk Factors for Pitcher Injuries" PI: S Powers, $100,000 — 2024–2026

**Major League Baseball Research Gift**
"Assessing Associations between Longitudinal Movement Screens and Performance in Baseball Pitchers"
PI: S Powers, $15,000 — 2026

### Internal

**Houston Methodist-Rice University Center for Human Performance Seed Grant Program**
"Utilizing Ultrasound-Based Shear Wave Elastography to Assess UCL Changes in Collegiate Pitchers"
PIs: S Powers, R Jack, $46,483 — 2026

**Rice University Creative Ventures Conference and Workshop Development**
"Rice Soccer Analytics Conference" PI: S Powers, $10,000 — 2024–2025

## Teaching

* indicates a course I designed

**Rice University**

| | |
|---|---|
| Instructor, *SMGT 430: Introduction to Sport Analytics | Spring 2024, 2026; Fall 2024, 2025 |
| Instructor, *SMGT 435: Baseball Analytics | Spring 2024; Fall 2024, 2025 |
| Instructor, STAT 699: Topics in Statistical Sciences | Spring 2026; Fall 2024, 2025 |
| Instructor, *SMGT 432: Soccer Analytics | Fall 2023 |
| Instructor, *SMGT 434: Football Analytics | Fall 2025 |
| Instructor, SMGT 373: Sport Analytics Internship | Spring 2026 |

**Stanford University**

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|---|---|
| Instructor, *STATS 50: Mathematics of Sports | Spring 2016 |
| TA, STATS 305A: Applied Statistics I | Fall 2016 |
| TA, STATS 216: Introduction to Statistical Learning | Winter 2014, Summer 2015, Winter 2017 |
| TA, STATS 202: Data Mining and Analysis | Fall 2013, Summer 2014 |
| TA, STATS 50: Mathematics of Sports | Fall 2014 |

**University of Chicago**

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|---|---|
| Instructor, STAT 23400: Statistical Models and Methods | Spring 2013 |
| TA, STAT 22000: Statistical Methods and Applications | Fall 2011, Spring 2012, Fall 2012 |

## Service

**To the Profession**

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|---|---|
| Lead Organizer, American Soccer Insights Summit | 2024– |
| Sports Column Co-Editor, *CHANCE* | 2026– |
| Associate Editor, *Journal of Quantitative Analysis in Sports* | 2026– |
| Associate Editor, *Journal of Statistics and Data Science in Sports* | 2026– |
| Associate Editor, SCORE Network | 2024– |
| Organizer, ASA Virtual Sports Analytics Conference | 2026 |

**Rice University**

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| Faculty Associate, Wiess College | 2024– |
| Instructor, STaRT@Rice (School of Social Sciences) | 2024, 2025 |
| Instructor, Owl Days Classroom Sampler (Office of Admission) | 2024 |
| Speaker, Special Occasion Dinner (Friends of Fondren Library) | 2024 |
| Speaker, Brains in a Bar (Association of Rice Alumni) | 2024 |
| Member, University Committee for Faculty and Staff Benefits | 2024 |

**Peer Review**

*Journal of Quantitative Analysis in Sports* (9), *Journal of Sports Analytics* (2), *International Journal of Sports Science & Coaching* (2), *Journal of Sport Management*, *Orthopaedic Journal of Sports Medicine*, *Journal of Sports Sciences*, *Experimental Physiology*, *Computer Methods in Biomechanics and Biomedical Engineering*, *Proceedings of the National Academy of Sciences*

## Professional Development

**Rice Center for Teaching Excellence**

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| INSTILL Mini-Grant Program | 2023–2025 |
| Reading Group | 2023–2025 |
| Symposium on Teaching and Learning | 2024 |
| Workshop on Inclusive Teaching Strategies | 2023 |

**Conferences Attended**

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|---|---|
| Women in Sports Data Symposium | Philadelphia 2023, 2024 |
| Minority Trailblazers in Sports Conference | Houston 2023 |

## Awards

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|---|---|
| Outstanding Associate, Wiess College | 2026 |
| Graduate Research Fellowship, National Science Foundation | 2011 |
| McCormick Fellowship, University of Chicago | 2011 |
| Mathematical Decision Sciences Award, UNC Department of Statistics & Operations Research | 2011 |
| Student Speaker Award, Pi Mu Epsilon National Meetings | 2010 |
| National Merit Scholarship, National Merit Scholarship Corporation | 2007 |
| Jack Kavanagh Memorial Youth Baseball Research Award, SABR | 2005, 2006, 2007 |

## Activities

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|---|---|
| Sports Analytics Club, Stanford University | 2014–2017 |
| Co-President | 2015–2016 |
| Intercollegiate Club Baseball, Stanford University | 2013–2015 |
| Intercollegiate Club Volleyball, Stanford University | 2013–2016 |

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|---|---|
| Intercollegiate Club Volleyball, University of Chicago | 2011–2013 |
| Intercollegiate Club Volleyball, University of North Carolina at Chapel Hill | 2007–2011 |
| President | 2008–2010 |